Nishal Stanislaus Silva, *PhD*

Machine Learning Research Engineer | Deep Learning, Time-Series, Pattern Detection | Applied ML

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

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Professional Summary

Machine learning engineer with 10+ years across industry and academia, delivering ML models that make live decisions under tight latency budgets. I build end-to-end: feature pipelines and training datasets, offline experiments, deployment into real-time and batch decision systems, and monitoring. Domains: financial risk and anomaly detection on structured data (Forestpin), real-time inference on streaming behavioral and sensor time-series, and production computer vision and IoT at manufacturing scale. Verified results include a real-time detector running at 14 ms with F1 0.76, 74× faster than the DTW baseline it replaced (JAES 2026), and industrial systems that improved operational efficiency by up to 300%. Published in JAES, IEEE and ACM venues; Claude Code is part of my day-to-day development workflow.

Core Competencies

ML Engineering: low-latency ML inference, classification and anomaly detection on structured (tabular) data, feature pipelines, training datasets, offline experiments and ablations, batch and real-time decision systems, model and data monitoring, sequence models (LSTM/GRU) for behavioral time-series
Domain Expertise: financial forensics, risk detection, compliance monitoring, anomaly scoring and triage, high-frequency IoT time-series, signal processing, industrial computer vision
Research & Leadership: benchmark and dataset design, evaluation metrics (F1, precision/recall, ROC-AUC, EER), code review and CI/CD, mentoring engineers, stakeholder requirements to technical specs, 10+ peer-reviewed papers

Technical Stack

Languages: Python, C++17, SQL, C, JavaScript/TypeScript, MATLAB, Shell
ML / AI: PyTorch, TensorFlow, Keras, scikit-learn, Pandas, NumPy, SciPy
Signal & Audio Processing: DSP, STFT, MFCC, Librosa, real-time streaming pipelines, Elk Audio OS, OSC
Data & Cloud: AWS (EC2, S3, ECS, MWAA, RDS), Snowflake, SQL, ETL pipelines
ML Lifecycle & MLOps: Airflow (training/ETL orchestration), MLflow, Docker, Jenkins, Git, pytest, CI/CD, REST APIs, Claude Code

Education

Ph.D - Information Engineering and Computer Science - *University of Trento, Italy* — *Jan 2025*
M.Sc - Telecommunication and Electronic Engineering - *Sheffield Hallam University, UK* — *Aug 2018*
B.Eng (Hons) - Electronic Engineering - *Sheffield Hallam University, UK* — *Mar 2014*

Professional Experience

Independent Research & Open-Source | *Self-directed, Toronto, Canada* — *Jan 2026 - Present*

- Benchmarked 6 scoring methods across 3 model families for a binary accept/reject classifier over 100,000 scored trials; best model reached ROC-AUC 0.822 (EER 0.267) vs. 0.599 for the baseline.
- Built an Airflow feature pipeline (melodiq): ingestion and feature-extraction DAGs, S3-to-Snowflake loading, Dockerized, infrastructure-as-code.
- Build all 2026 projects with AI-assisted workflows in Claude Code; maintain Nebula, an open-source C++17 feature library; two papers accepted (IEEE IS2 2026, JAES).

Postdoctoral Researcher | *University of Trento, Trento, Italy* *Jan 2025 - Dec 2025*

- Owned the full ML pipeline for streaming audio and IMU/gesture data on the EU Horizon EIC Pathfinder project MUSMET: acquisition, feature engineering, training and evaluation, low-latency deployment, and monitoring.
- Took a real-time detector from prototype to deployment: 14 ms inference on Raspberry Pi 4 at F1 0.76 and 31.4% CPU, 74× faster than the 1038 ms DTW baseline (JAES 2026); inference services held under 30 ms.
- Designed offline benchmark suites with baselines and ablations (RNN vs. DTW, audio vs. MIDI, pattern-set sizes 1/3/10) to quantify accuracy, latency and CPU trade-offs before choosing what ships.

Visiting Researcher | *McGill University (IDMIL / CIRMMT), Montreal, Canada* *May 2024 - Aug 2024*

- Built real-time behavioral gesture detection (LSTM on a 1 s IMU window) fused with audio pattern detection via a hierarchical state machine; P 0.79 / R 0.76 / F1 0.78 on a 500-event corpus (IEEE IS2 2025).

Visiting Researcher | *University of Visual and Performing Arts, Colombo, Sri Lanka* *Jan 2024 - Mar 2024*

- Human-centred evaluation of ML-driven interactive systems: metrics plus user interviews (IEEE I3DA 2025).

Technical Consultant | *Forestpin (Pvt) Ltd, Colombo, Sri Lanka* *Aug 2020 - Feb 2021*

- Designed and deployed ML and statistical pipelines for financial forensics, compliance monitoring and risk detection on large structured enterprise datasets.
- Built SQL + Python backend scoring services for anomaly flagging and triage in production.
- Translated compliance-stakeholder requirements into technical specifications for the modelling work.

Research Engineer | *MAS Holdings (Pvt) Ltd, Colombo, Sri Lanka* *Jan 2015 - Jul 2020*

- Delivered end-to-end ML and computer-vision systems for manufacturing at scale, integrating camera, sensor and IoT streams into production workflows; improved operational efficiency by up to 300%. Mentored engineers and introduced CI/CD and code review.
- Built real-time ingestion and ETL for high-frequency IoT and operational time-series across sites in South Asia, North America and Africa, with C++/Python inference engines on embedded hardware.
- Automated care-label quality control as a live accept/reject decision system (OpenCV matching, conveyor with PLC reject), cutting inspection time by 99.5%.

Selected Publications (full list: *nishal.xyz/publications*)

- Silva, N. and Turchet, L. *Real-Time Audio Pattern Detection for Smart Musical Instruments*. Journal of the Audio Engineering Society, Vol. 74, No. 3, Mar. 2026.
- Silva, N. and Turchet, L. *Smart Drums: Comparing Audio and MIDI in Embedded Real-Time Drum Pattern Recognition*. Journal of the Audio Engineering Society, 2026 (in press).
- Silva, N. and Turchet, L. *Music Information Retrieval as a Service: Latency Characterization of an Edge-to-Server Architecture for Near-Real-Time Audio Analysis*. IEEE International Symposium on the Internet of Sounds, 2026 (accepted).

Independent Research, Highlights, & Recognition

- **MIDI Innovation Awards 2023:** Finalist, "Hot Licks" real-time pattern detection for smart instruments.
- **GMOA Recognition (Sri Lanka):** COVID-19 vitals-monitoring CV device deployed across hospital wards.
- **Open datasets:** Four open-access benchmark datasets on Zenodo, 9,800+ recordings from 70+ musicians.